

Probability Is All You Need

- History
- Descriptive Statistics
- Definition of P
- P and Distribution, Expectation, Variance
- P and Central Limit Theorem
- P and Hypothesis Test
- P and Bayes
- P and Stochastic Process

History

The birth of statistics

- Math: Pascal and Fermat's researches on gambling. Since 1654.
- Physics: Einstein, Perrin, Langevin, Ito's researches on Brownian Motion. Since [1905](#).
- Biology: Galton, Pearson, Fisher's researches on Eugenics. Since [1883](#).
- NLP: Markov's researches on Pushkin's poem *Eugene Onegin*. Since [1913](#).

In the math part, Bernoulli, De Moivre, Laplace, Poisson also made great contributions. In 1900, Hilbert's sixth question is about the axiomatization of probability and physics. In 1933, Kolmogorov did that.

But the true birth of statistics relied on the latter 3 parts, especially biology. **From the birth, statistics has deep connections with biology, physics and NLP.**

Descriptive Statistics

In primary school, junior high school and senior high school, we learnt a lot about the so-called "descriptive statistics".

For example

- boy-girl-ratio: [France](#). [China](#).
- basketball, soccer: see [kaggle](#).
- weighted average: like grades weighted by points.

But we can't just do this kind of thing, we must go beyond, build a model. Sometimes people call this "statistical inference". Here are 3 important things we need to do.

- population: just like the population of a country.
- sample: both a number and a random variable.
- $f \rightarrow p$

Definition of P

- Just like throwing a coin
- Measure

P and Distribution, Expectation, Variance

- Distribution: Discrete and Continuous
- E, Var: See them as Operator!
 - $E(b) = E_x(E_b(b|x))$
 - $Var(b) = E_x(Var_b(b|x)) + Var_x(E_b(b|x))$
 - [MSE = Bias + Variance + Noise](#)
 - $E(\frac{1}{n-1} \sum_{i=1}^n (X_i - \text{mean of the data})^2) = Var(X)$

Example from RL: remember the slot machine in class?

I must emphasize that the distribution has more information than E, Var. That means, when you get a data in your project, first plot its distribution!

P and Central Limit Theorem

[Wikipedia.](#)

In the simplest form,

$$\forall i, X_i \text{ i.i.d.} \Rightarrow \sum_{i=1}^n X_i \xrightarrow{d} N(nE(X_i), nVar(X_i))$$

And this is the true reason why Gauss choose e^{-x^2} instead of $\cos(x)$, e^{-x^4} , $(\frac{1}{1+e^{-x}})'$, $(\arctan(x))'$ and so on.

There is an example from neuroscience, which I will call the core of biological neural network. Based on Central Limit Theorem, a neuron in a big neural network should have $\frac{\sigma}{\mu} \rightarrow 0$, as $n \rightarrow \infty$, but this contradicts with the experiment result, which show finite $\frac{\sigma}{\mu}$.

There are 2 solutions

- balanced exhibition and inhibition
- correlated inputs

P and Hypothesis Test

[Lady Tasting Tea](#)

Other hypothesis test: χ^2 , t, F, permutation test, rank, KS, ...

Further Reading

- [你是否相信十二星座中的某些星座更适合在一起?](#)
- [The Misuse of p-value in the Senior High School](#)
- [p-hacking: What it is and how to avoid it!](#)
- Example from Neuroscience: [Sen Song's work](#)

P and Bayes

- Hypothesis Test
- [Three door problem](#)
- Three choice problem

P and Stochastic Process

- Markov Process
- Poisson Process
- Steady Process